Topic 6

Memory

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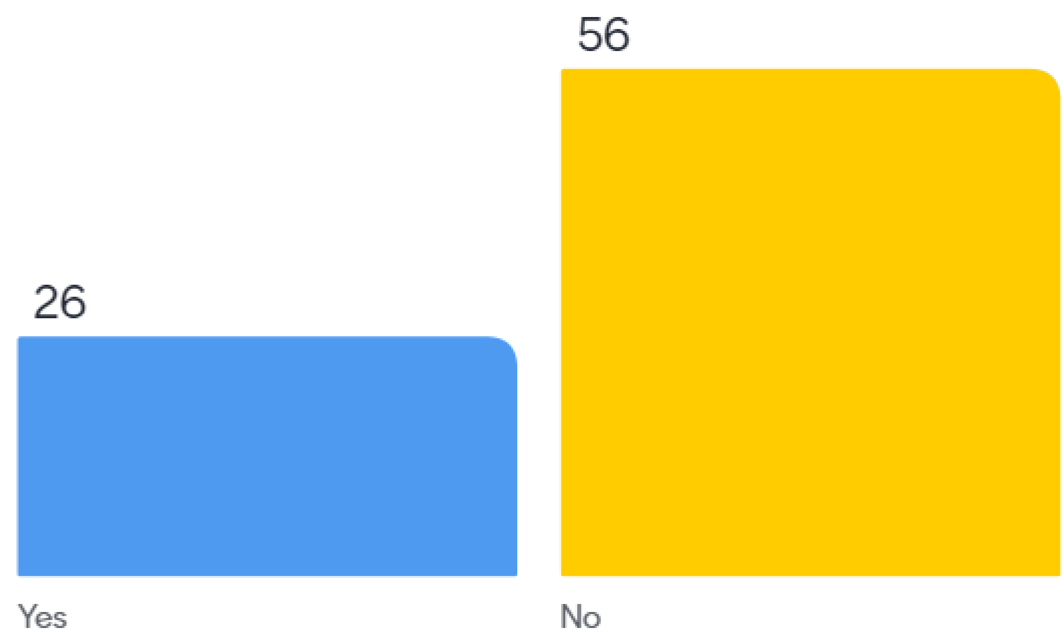
Do you believe you can remember events from your past accurately?



Overview

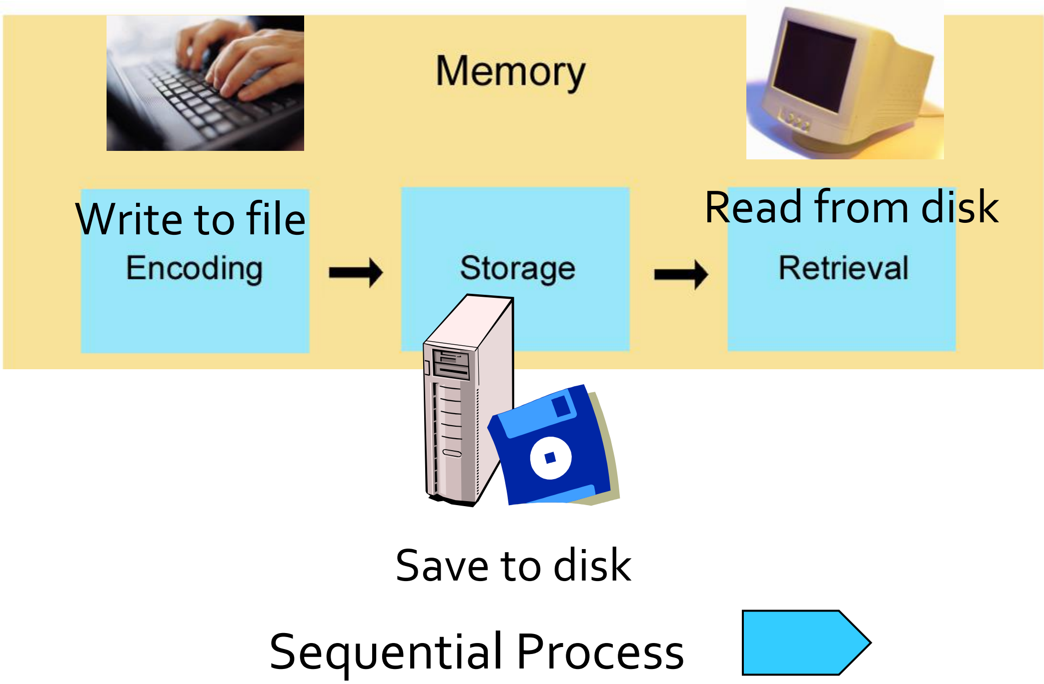
- The study of memory
- Encoding
- Storage
- Retrieval
- Varieties of memory
- When memory fails

The study of memory

- Perception
  - the organization of current stimuli
- Memory
  - The organization of stimuli from the past (ideas and events)
  - The nervous system's capacity to retain and retrieve skills and knowledge
  - We have multiple memory systems, and each memory system has its own "rules"

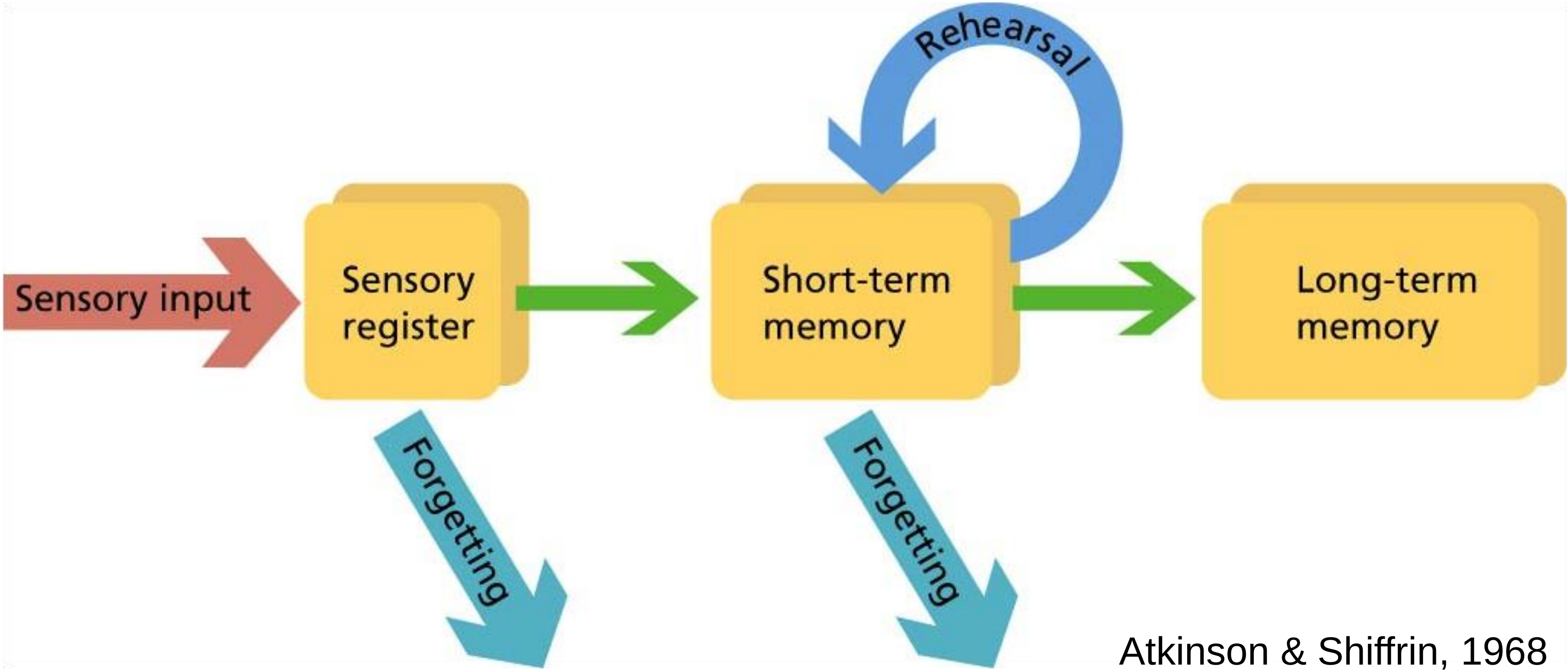
The study of memory

- Memory operates over time in three phases
  - Encoding
  - Storage and consolidation (in memory traces)
  - Retrieval
    - recall
    - recognition



Encoding

- Three memory systems: stage theory of memory



Atkinson & Shiffrin, 1968

Sensory Memory

- Sensory memory feeds our active working memory, recording momentary images, sounds, and strong scents.
- But sensory memory, like a lightning flash, is fleeting.

Sensory Memory

- **Iconic memory**
  - A momentary sensory memory of visual stimuli; a photographic or picture-image memory lasting no more than a few tenths of a second.

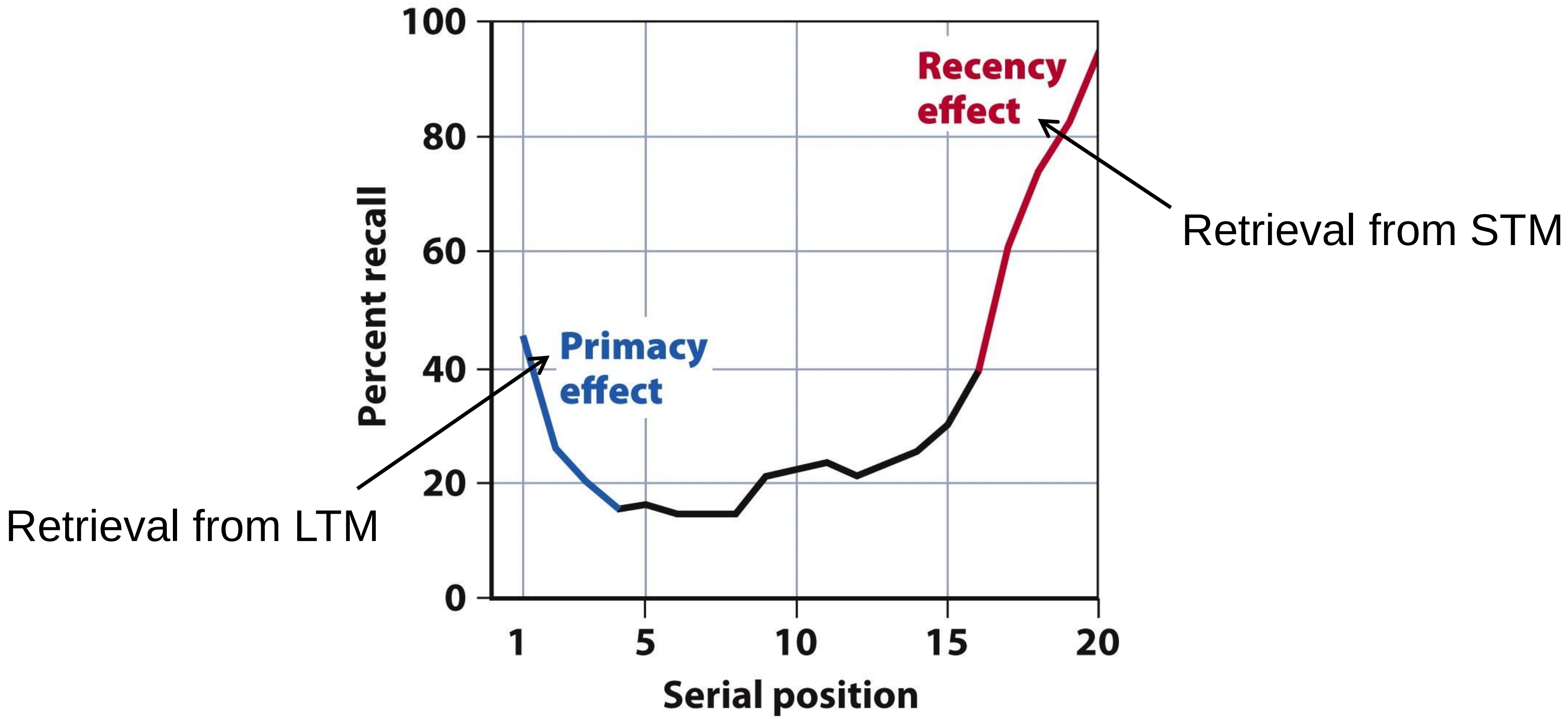
- **Echoic memory**
  - A momentary sensory memory of auditory stimuli; if attention is elsewhere, sounds and words can still be recalled within 3 or 4 seconds.

Encoding

- STM
  - Limited capacity
    - Memory span task: magic number seven
      - Items disappear as a result of decay and replacement
    - By recoding (chunks) -> expansion of capacity
      - e.g., UKEUCIAPHDFBIUSSR
      - e.g., UK EU CIA PHD FBI USSR
- LTM
  - Unlimited capacity
- Through rehearsal STM -> LTM

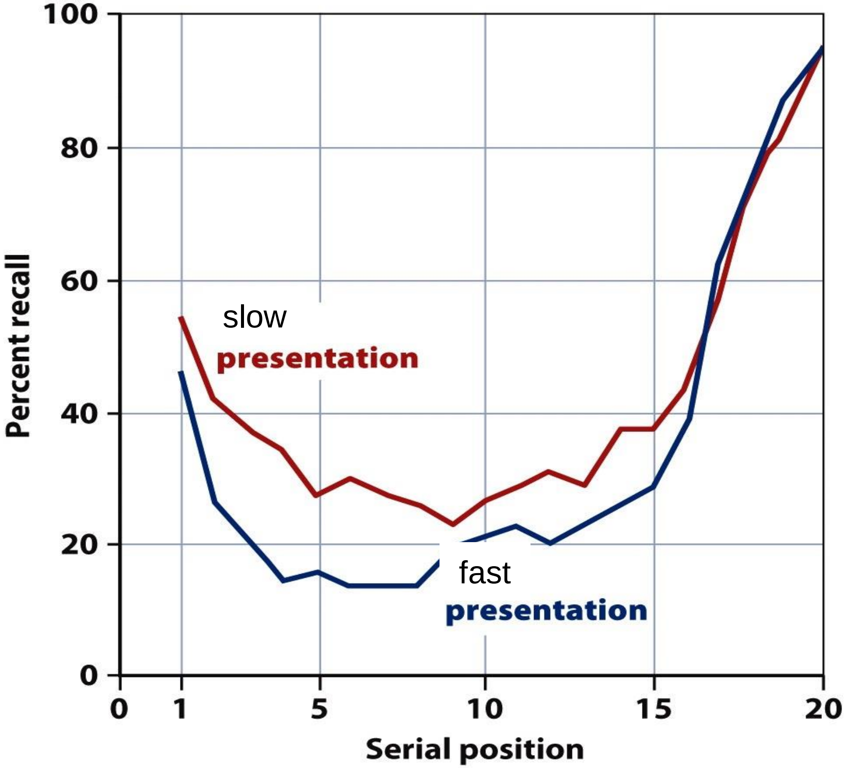
Encoding

- Evidence for two systems (STM-LTM)
  - Free-recall task



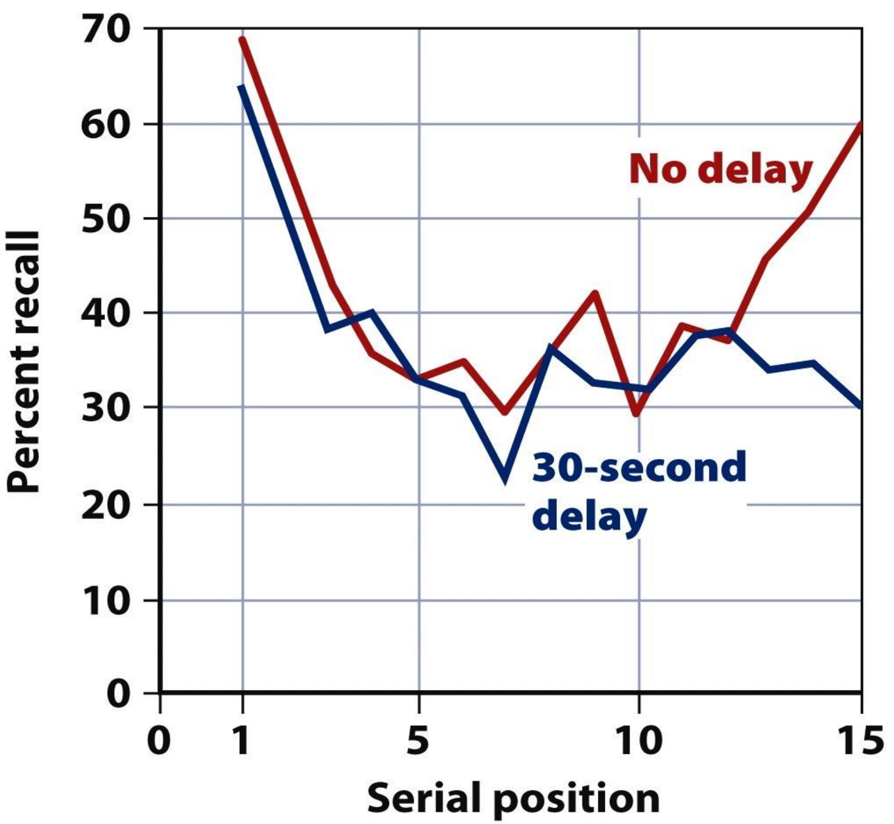
Encoding

- Evidence for two systems (STM-LTM)
  - Free-recall task -> faster presentation



Encoding

- Evidence for two systems (STM-LTM)
  - Free-recall task -> delay in recall



Encoding

- Rehearsal: STM -> LTM

- Experiment Craik & Watkins (1973)
- Experiment Nickerson & Adams (1979)

Encoding

Daughter
Oil
Gras
Grandmother
Father
Car
Science
etc..

Report the most recent word starting with the letter "g"

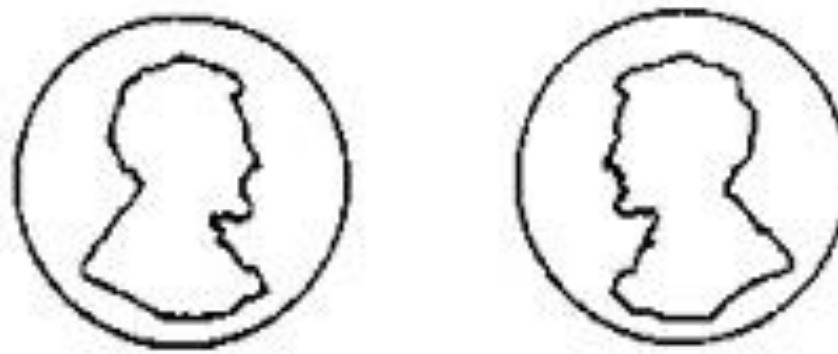
Recall-test:
Recall as many words starting with the letter "g" as possible

Result: g-words were not better remembered than other words -> time in STM does not determine p(recall)

Craik & Watkins (1973)

Encoding

Does the Lincoln
on the penny face
right or left?



Nickerson & Adams (1979)

Encoding

Nickerson & Adams (1979)

Encoding

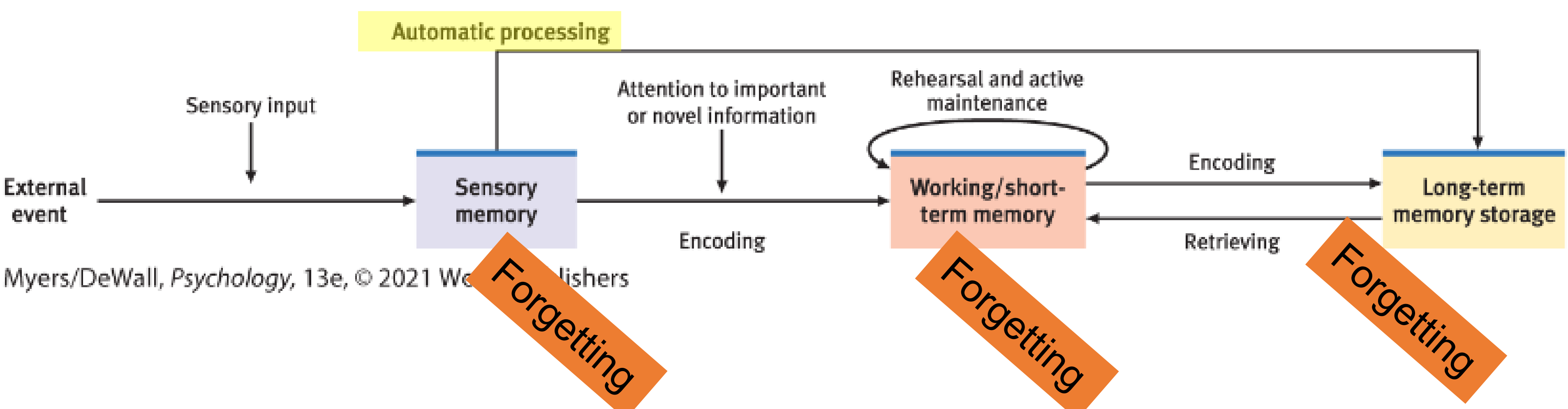
- Rehearsal: STM -> LTM

- Experiment Craig & Watkins (1973)
- Experiment Nickerson & Adams (1979)

- Memory is an active process requiring attention

Encoding

- Modern memory theory
  - STM -> working memory
    - Information one is thinking about
  - Long-term memory (LTM)
    - Information, stored during longer intervals, which is currently not actively used.



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Encoding

- Which encoding processes promote later recall?
  - Depth of processing
    - Shallow processing (surface form)
    - Deep processing (meaning)

Sample Questions to Elicit Different Levels of Processing	Word Flashed	Yes	No
Most shallow: Is the word in capital letters?	CHAIR	______	______
Shallow: Does the word rhyme with train?	brain	______	______
Deep: Would the word fit in this sentence? The girl put the ______ on the table.	puzzle	______	______

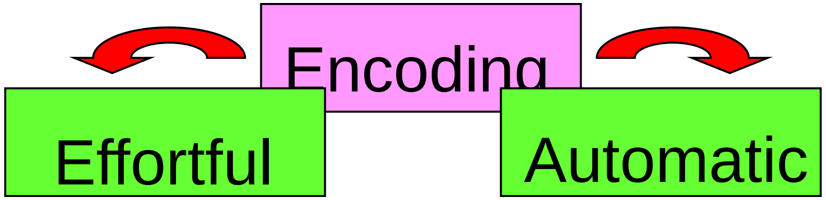
  - Understanding
    - A **schema** may help you to understand and subsequently remember information contents

Encoding

Good encoding provides an effective means for retrieval later.

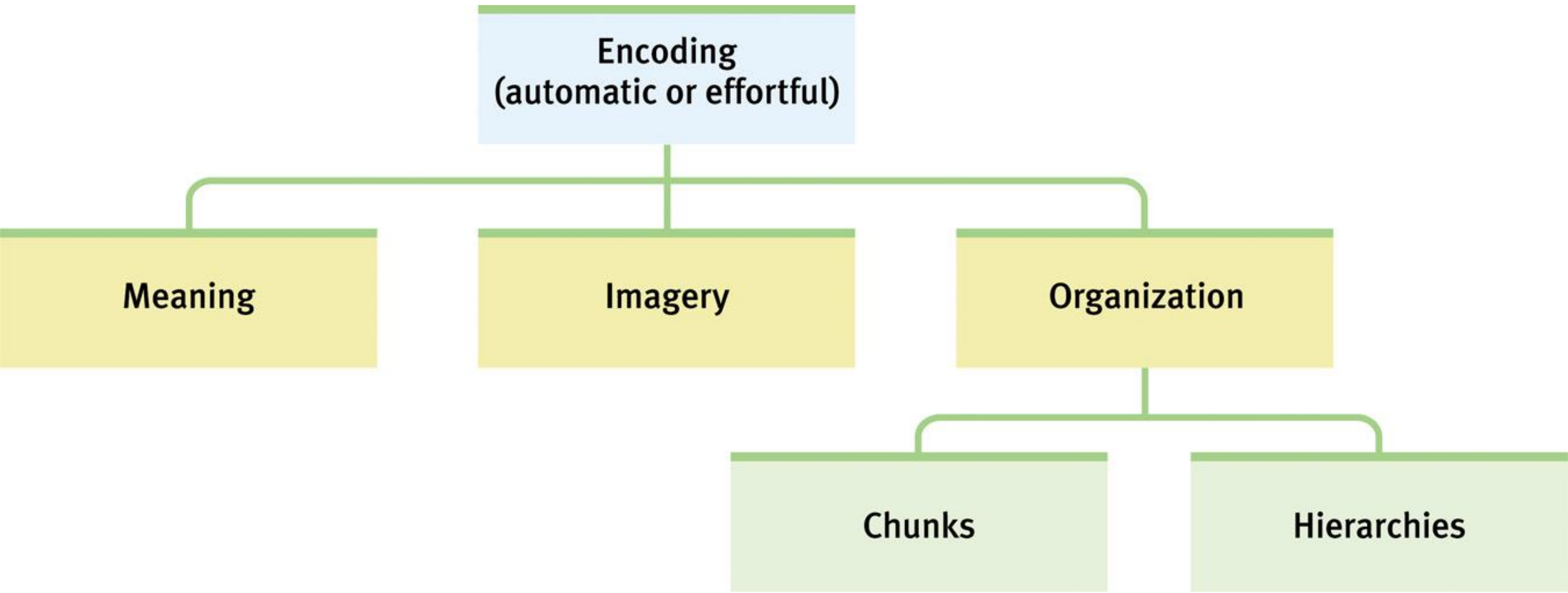
Encoding: Getting Information In

- **Effortful processing**
  - Requires attention and conscious effort.
    - Your new pinpass code



- **Automatic processing**
  - Unconscious encoding of incidental information
    - space, time, and frequency
  - and of familiar or well-learned information
    - sounds, smells, and word meanings
    - route to school

Encoding Summarized in a Hierarchy



Encoding by meaning

Example: patella

(kneecap)

- Encoding meaning (semantic encoding) results in better recognition later than visual or acoustic encoding

Visual Encoding

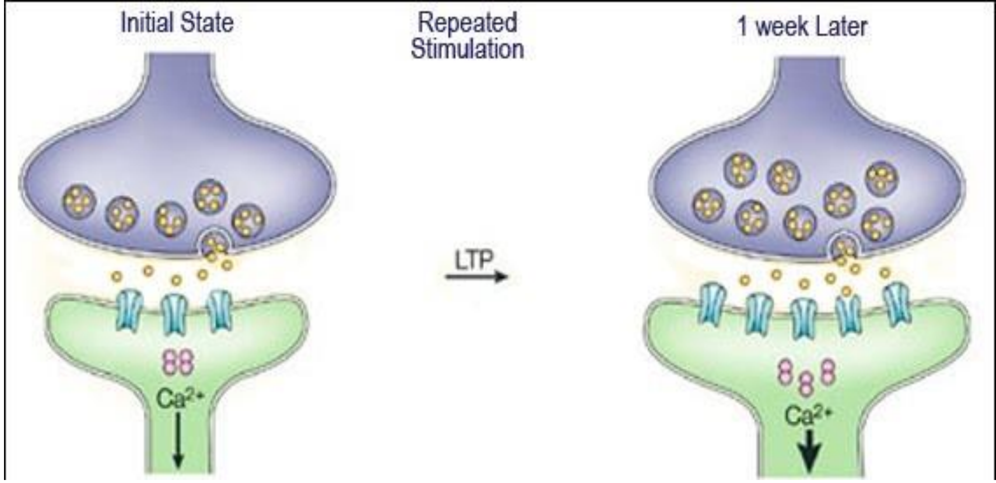
Mental pictures (imagery) are a powerful aid to effortful processing, especially when combined with semantic encoding.

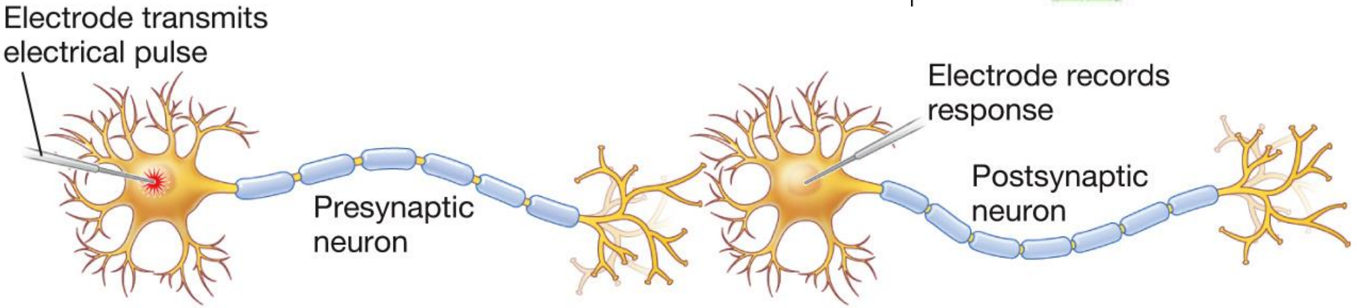
Showing anorexia in a picture may be more powerful than simply talking about it.

Storage

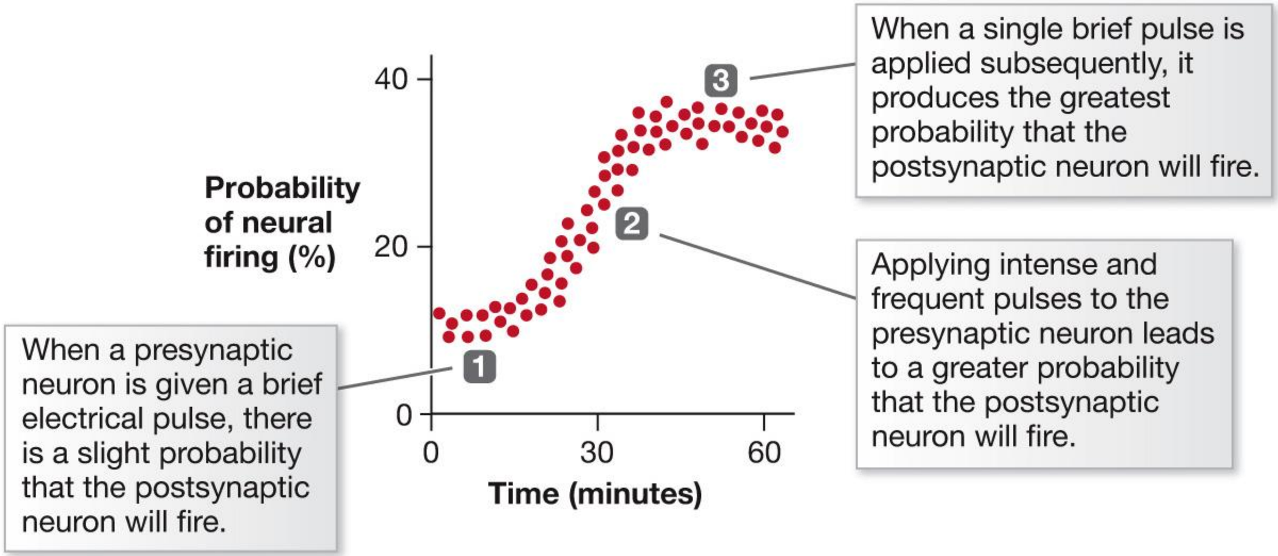
- After encoding -> storage in memory traces
  - Memory consolidation
    - the neural process by which encoded information becomes stored in memory -> transient and fragile potential memory contents are transformed into a more permanent state
    - **Long-term potentiation (LTP)**:
      - strengthening of a synaptic connection, making the postsynaptic neurons more easily activated by presynaptic neurons

Storage


Initial State
Repeated Stimulation
1 week Later
LTP
Ca^{2+}
Ca^{2+}

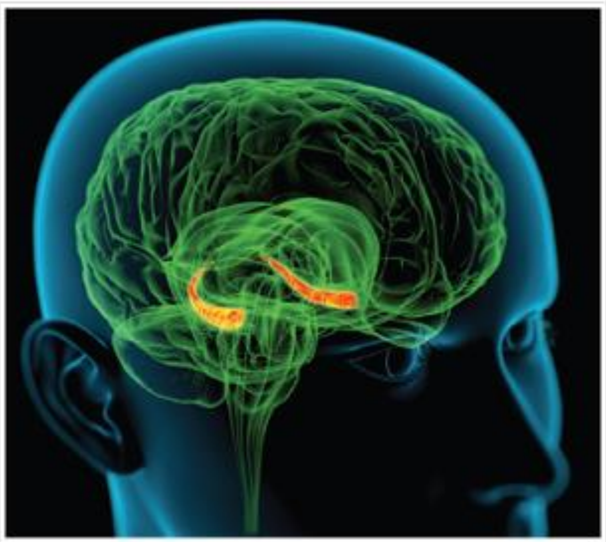

Electrode transmits electrical pulse
Presynaptic neuron
Electrode records response
Postsynaptic neuron

(b)


Probability of neural firing (%)
40
20
0
0
30
60
Time (minutes)
1
2
3
When a presynaptic neuron is given a brief electrical pulse, there is a slight probability that the postsynaptic neuron will fire.
Applying intense and frequent pulses to the presynaptic neuron leads to a greater probability that the postsynaptic neuron will fire.
When a single brief pulse is applied subsequently, it produces the greatest probability that the postsynaptic neuron will fire.

Storage

- After encoding -> storage in memory traces
  - Where?



Roger Harris/Science Source

FIGURE 8.8 The hippocampus Explicit memories for facts and episodes are processed in the hippocampus (orange structures) and fed to other brain regions for storage.

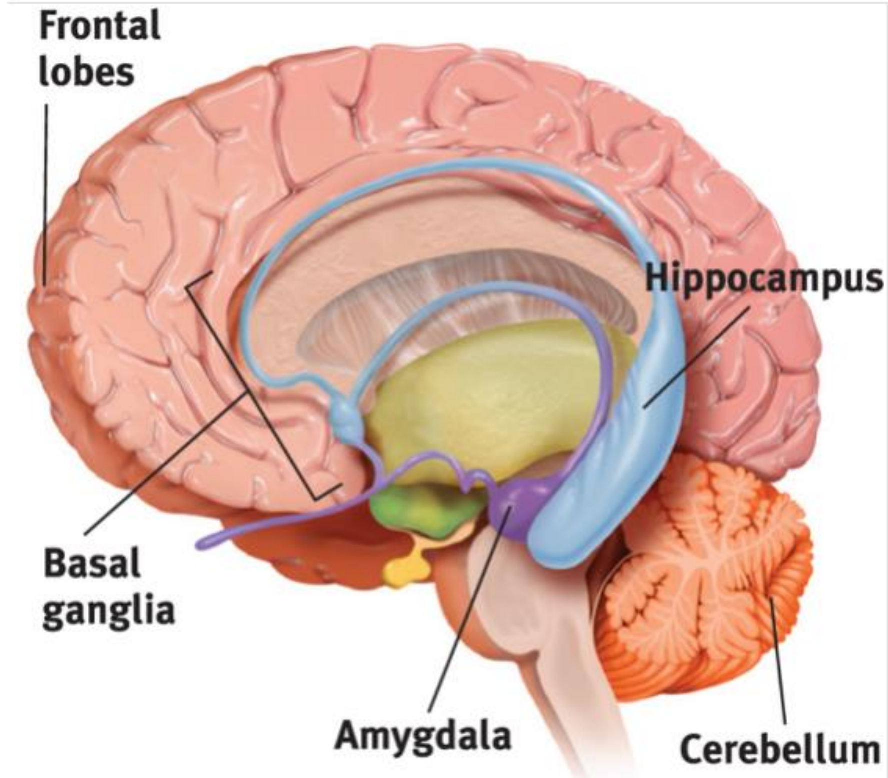


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FIGURE 8.9 Review key memory structures in the brain

Frontal lobes and *hippocampus:* explicit memory formation

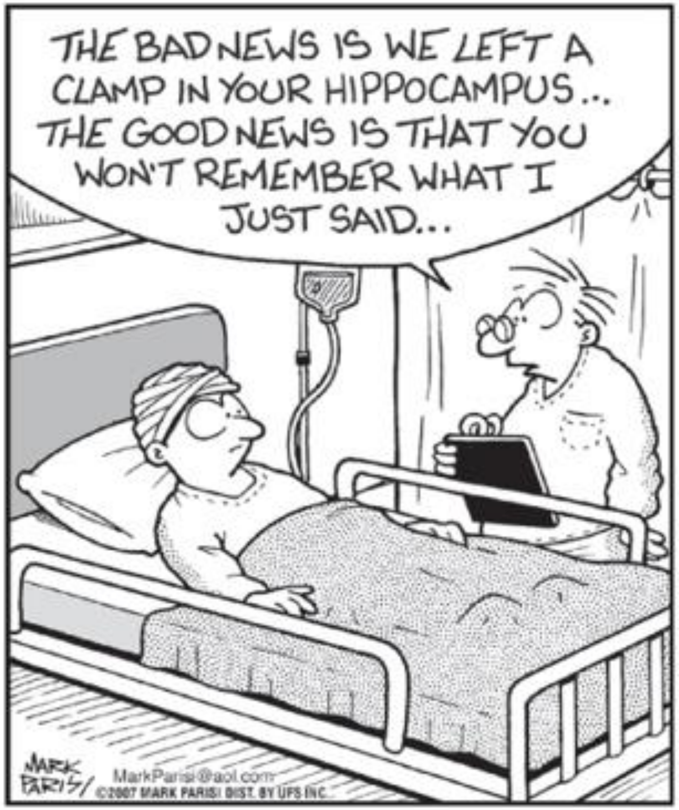
Cerebellum and *basal ganglia:* implicit memory formation

Amygdala: emotion-related memory formation

Damage to hippocampus

- Disrupts the formation and recall of explicit memories.

- **Memory consolidation**
  - The neural storage of a long-term memory.
  - Sleep supports this.



Mark Parisi/offthemark.com

The Amygdala, Emotions, and Memory

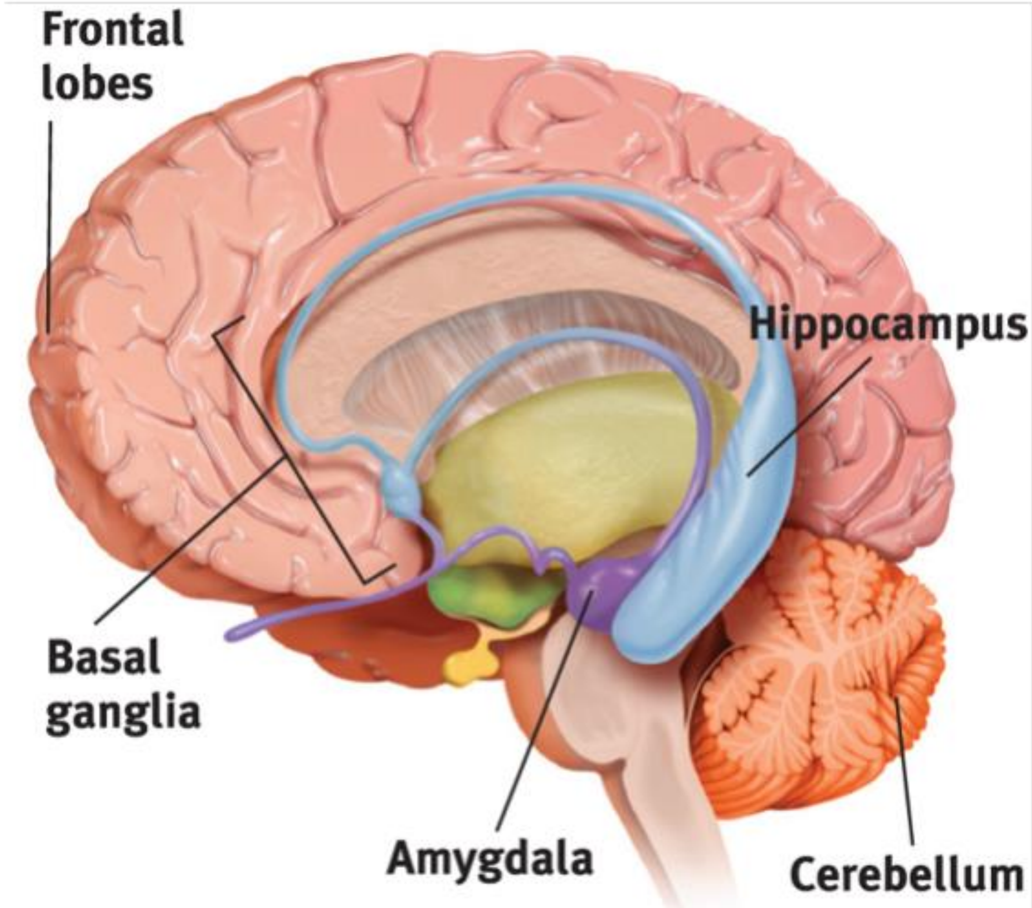


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FIGURE 8.9 Review key memory structures in the brain

Frontal lobes and *hippocampus:* explicit memory formation

Cerebellum and *basal ganglia:* implicit memory formation

Amygdala: emotion-related memory formation

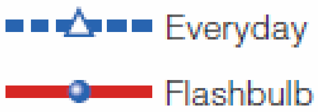
Stress Hormones & Memory

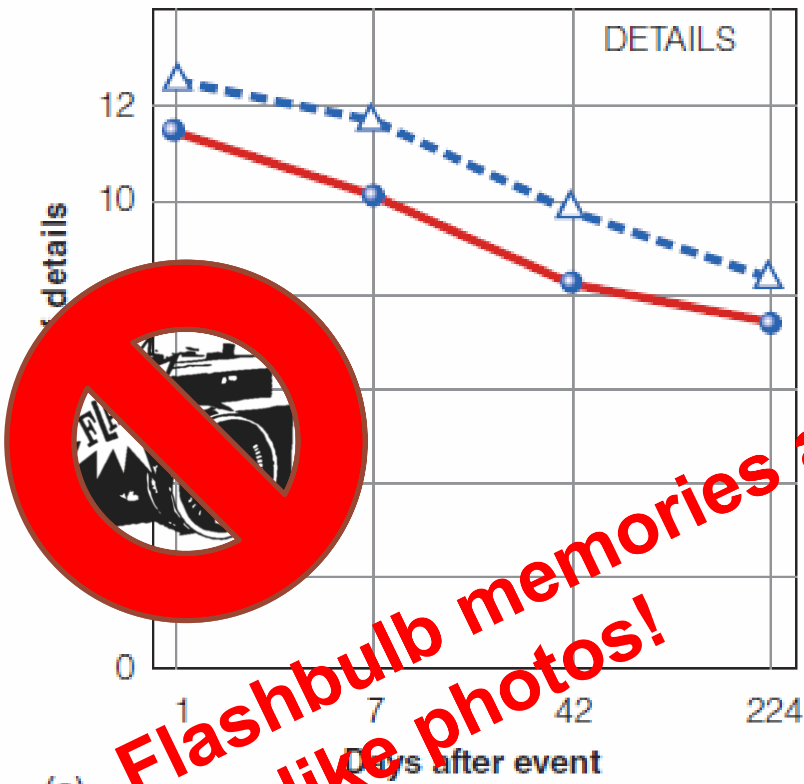
- Heightened emotions (stress-related or otherwise) make for stronger memories.

- **Flashbulb memories** are clear memories of what we were doing during emotionally significant moments or events.

- Stress hormones make more glucose energy available to fuel brain activity

Flashbulb memories about 9/11





(a)

Flashbulb memories are <u>not</u> like photos!

Memories for flashbulb events decay (just as for regular events)!

Talarico & Rubin (2003)

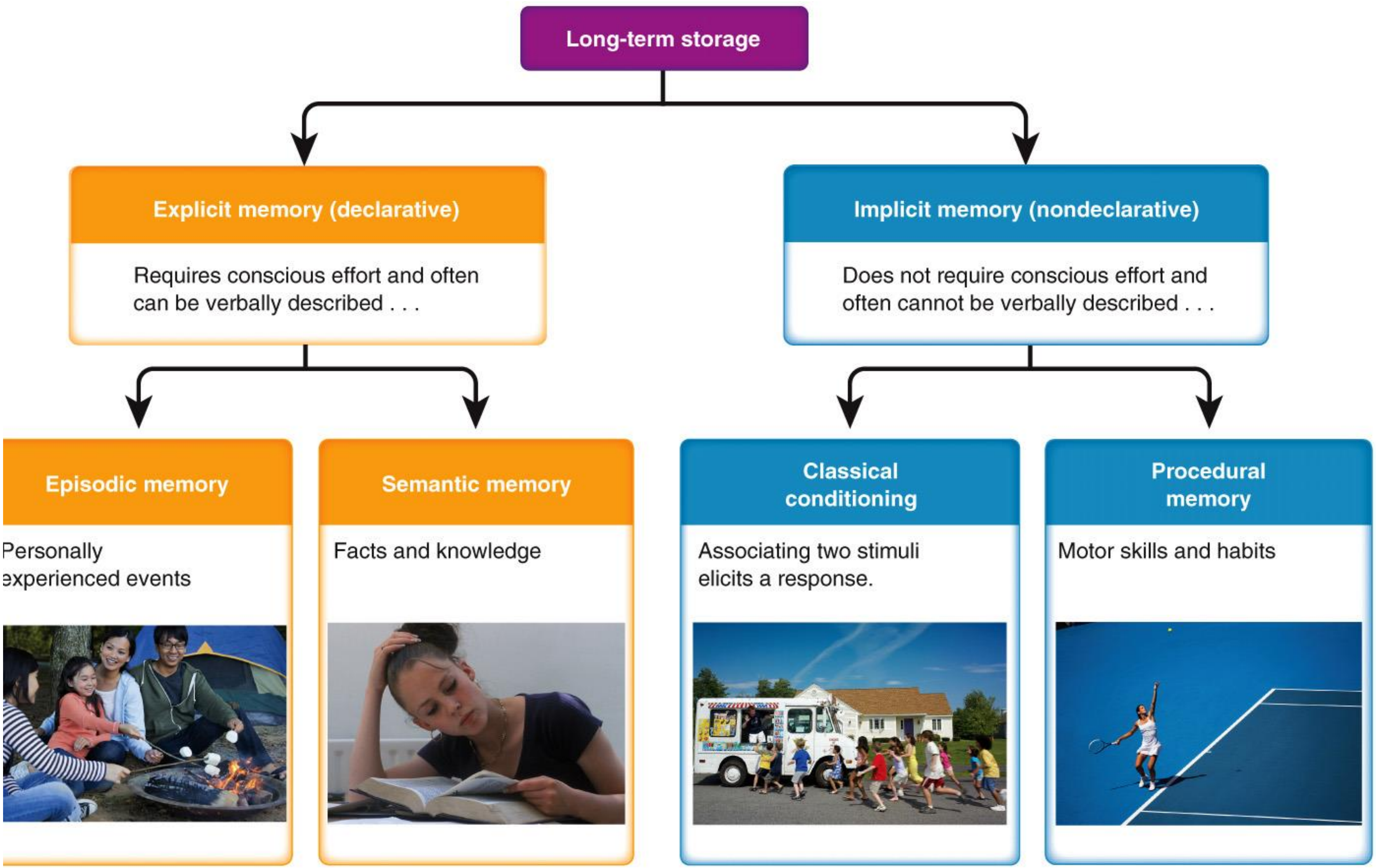
Flashbulb memories about

Talarico & Rubin (2003)

Storage

- After encoding -> storage in memory traces
  - Memory consolidation
    - the neural process by which encoded information becomes stored in memory -> transient and fragile potential memory contents are transformed into a more permanent state
    - **Long-term potentiation (LTP)**: strengthening of a synaptic connection, making the postsynaptic neurons more easily activated by presynaptic neurons
    - After rehearsal -> **reconsolidation** (might corrupt)

Varieties of memory



- **Explicit memory**
  - Retention of facts and experiences that we can consciously know and “declare.”
  - Declarative memory

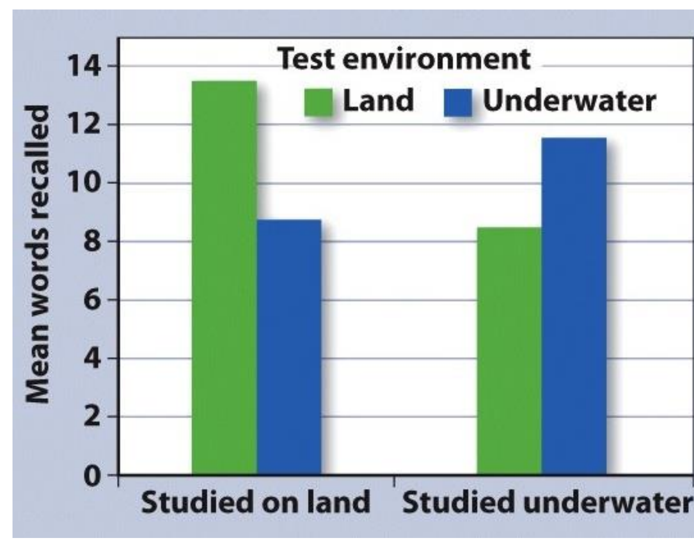
- **Implicit memory**
  - Retention of learned skills or classically conditioned associations independent of conscious recollection.
  - Nondeclarative memory

Retrieval

- Recognition
  - The person must identify an item amongst other choices.
    - A multiple-choice test requires recognition
- Recall
  - the person must retrieve information using effort
    - A fill-in-the blank test requires recall.

Retrieval

- Good retrieval cues
  - re-create the context in which the original learning occurred -> context reinstatement
    - e.g., experiment divers
  - re-create the state of mind in which the original learning occurred



Godden and Baddeley (1975)

Retrieval

- Good retrieval cues
  - re-create the context in which the original learning occurred -> context reinstatement
    - e.g., experiment divers
  - re-create the state of mind in which the original learning occurred
  - A richer encoding context results in better recall.

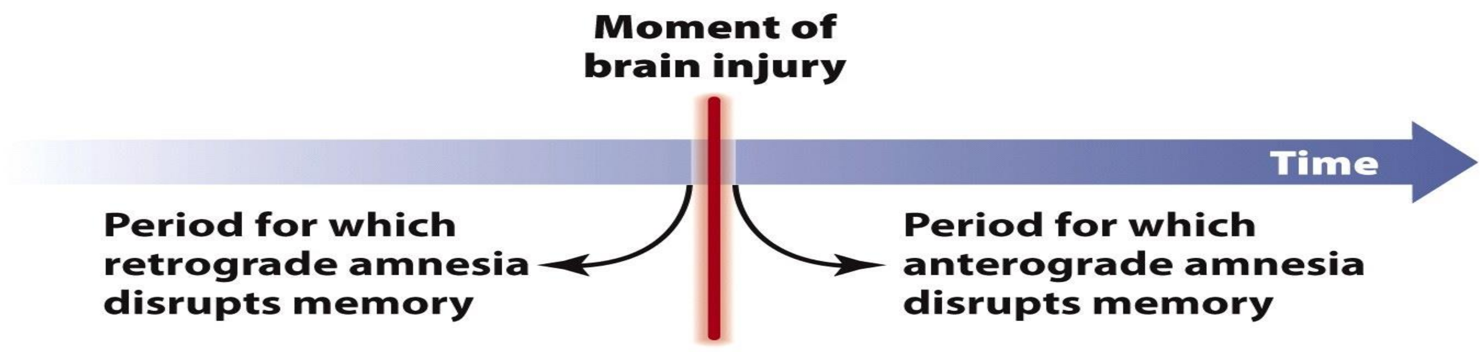
→ **Encoding specificity principle**

Forgetting

- An inability to retrieve information due to poor encoding, storage, or retrieval.

When memory fails

- **Amnesia**: a deficit in long-term memory—resulting from disease, brain injury, or psychological trauma—in which the individual loses the ability to retrieve vast quantities of information.

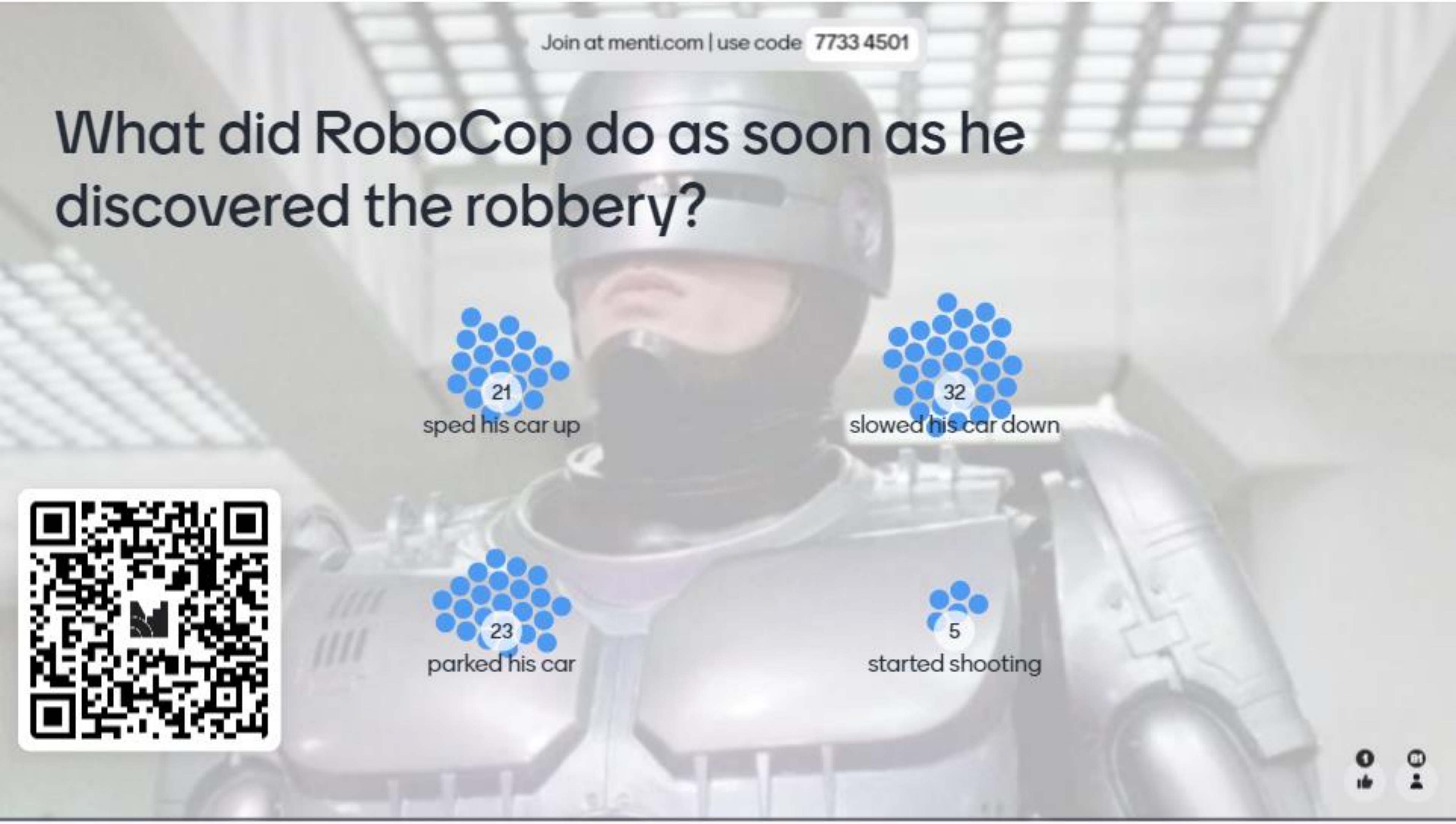


When memory fails

- An inability to remember
  - Encoding Failure
    - We cannot remember what we do not encode. Information never enters long-term memory.
  - e.g., absentmindedness, memory decay
- Memory reconstructions (distortions of memory)
  - e.g., bias, misattribution, suggestibility

When memory fails

- **Absentmindedness**:
  - the inattentive or shallow encoding of events
    - During most of our daily activities we are consciously aware of only a small portion of both our thoughts and our behaviors.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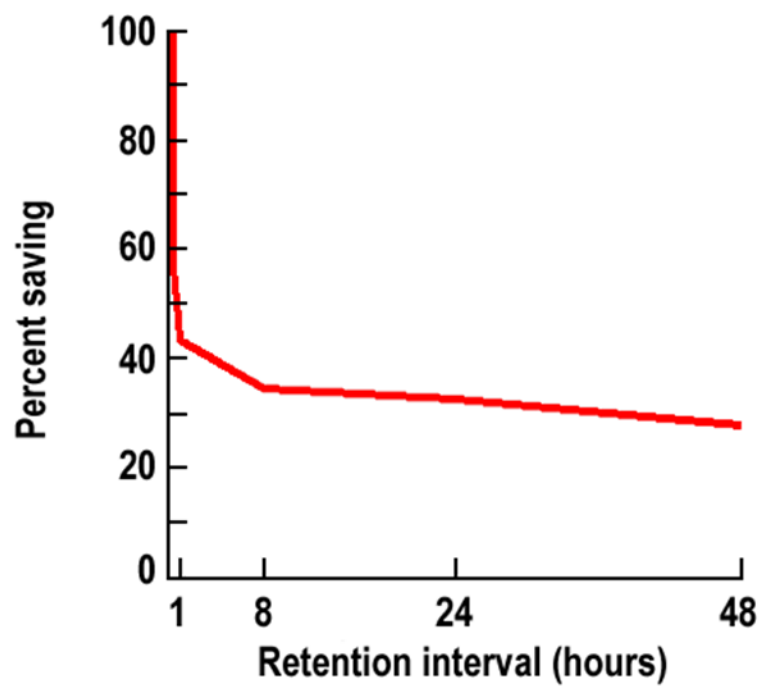
What did RoboCop do as soon as he discovered the robbery?
21
sped his car up
32
slowed his car down
23
parked his car
5
started shooting

When memory fails

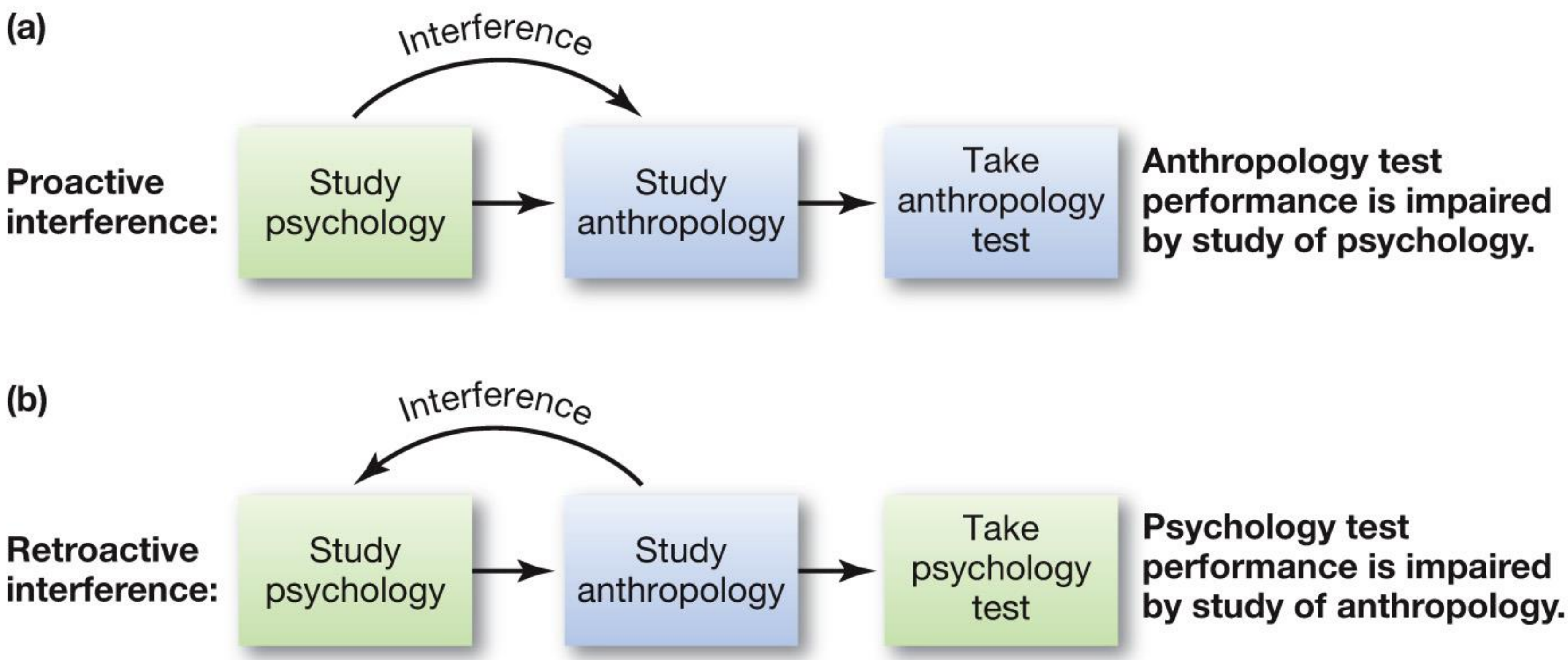
- **Memory decay**: forgetting over time
  - Forgetting curve (Ebbinghaus, 1885)



When memory fails

- **Memory decay**: forgetting over time
  - Forgetting curve (Ebbinghaus, 1885)
  - Why?
    - Decay of memory traces with time (caused by metabolic processes)
    - Interference
      - After sleep less forgetting than after wakefulness
      - Proactive and retroactive interference

When memory fails



When memory fails

- **Memory bias**:
  - the changing of memories over time so that they become consistent with current beliefs or attitudes
  - flashbulb memories can be biased

When memory fails

- **Source misattribution**:
  - memory distortion that occurs when people misremember the time, place, person, or circumstances involved with a memory.

When memory fails

Mice run on average 6 km/hour.

Crocodiles sleep with open eyes.

etc.

How interesting is this to you?

Mice run on average 6 km/hour.

Children eat on average 4 kilo peanut butter a year.

etc.

How likely is this true?

When memory fails

- **Suggestibility**:
  - the development of biased memories from misleading information
    - Different wordings of questions after an event may alter the participants’ memories of the event **->** Loftus

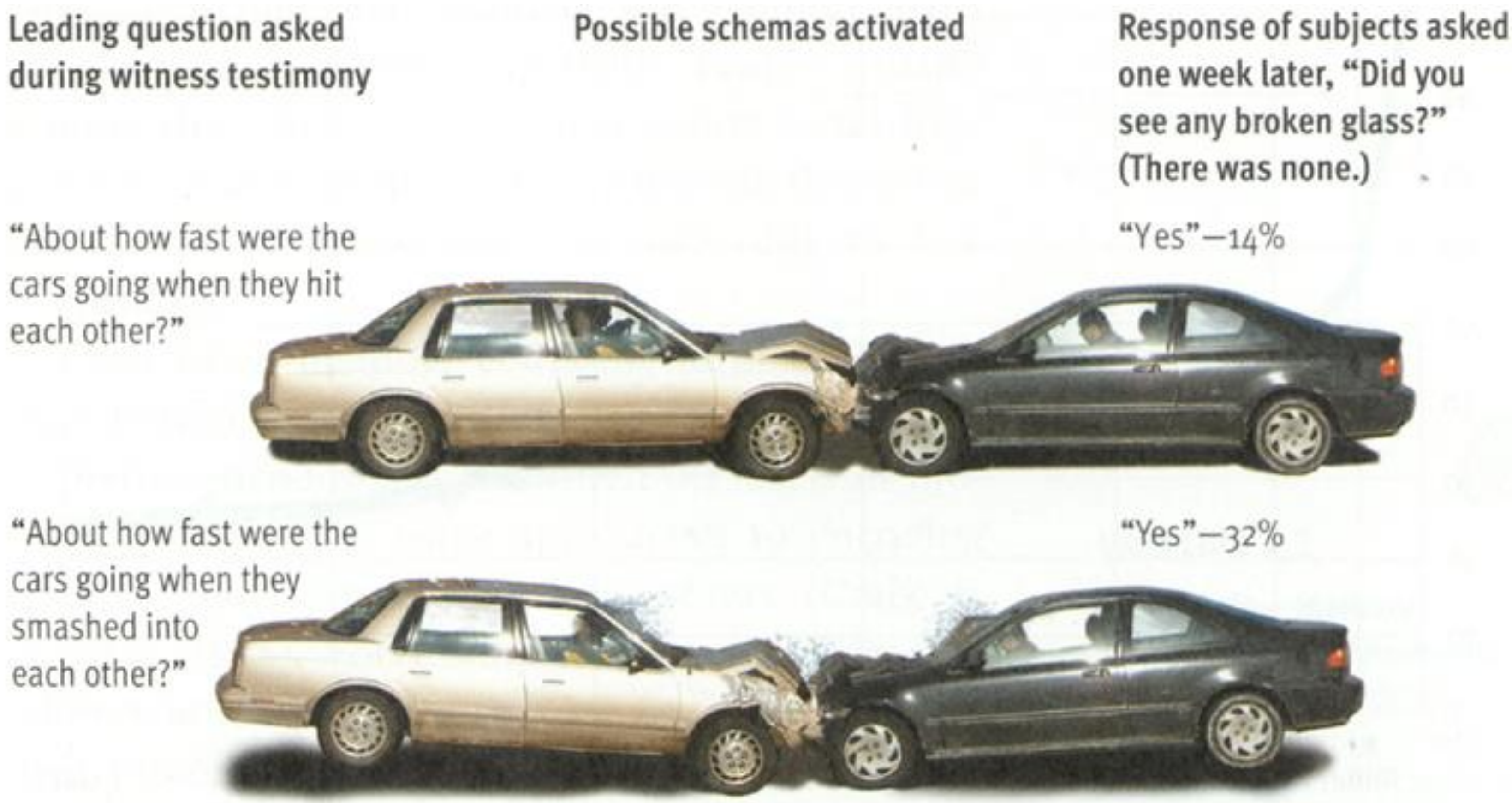
Memory Construction

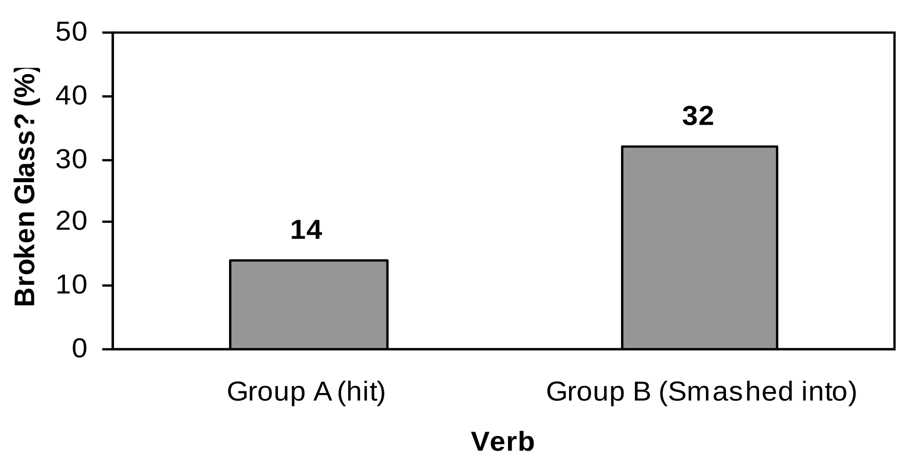
While tapping our memories, we filter or fill in missing pieces of information to make our recall more coherent.

Misinformation Effect: Incorporating misleading information into one's memory of an event.

After exposure to subtle misinformation, many people misremember.

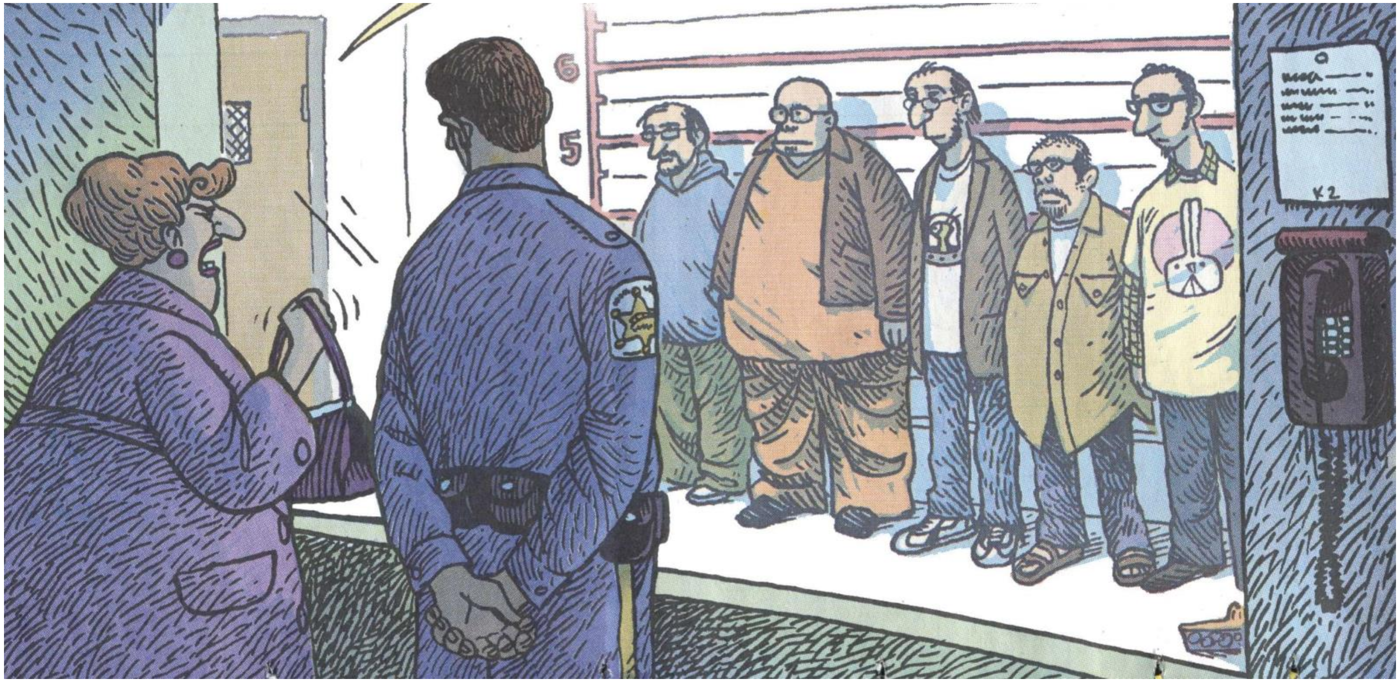
Guessed details are absorbed into our memories.


Leading question asked during witness testimony
Possible schemas activated
Response of subjects asked one week later, "Did you see any broken glass?" (There was none.)
"About how fast were the cars going when they hit each other?"
"Yes"—14%
"About how fast were the cars going when they smashed into each other?"
"Yes"—32%


50
40
30
20
10
0
Broken Glass? (%)
14
32
Group A (hit)
Group B (Smashed into)
Verb

When memory fails

- Incorrect eyewitness statements can be just as detailed, emotional and certain as correct statements.



When memory fails

- **Suggestibility**:
  - the development of biased memories from misleading information
    - Different wordings of questions after an event may alter the participants' memories of the event **->** Loftus
  - Memories can be distorted, or even implanted, with false information

Children’s Eyewitness Recall

Children’s eyewitness recall can be unreliable if leading questions are posed.

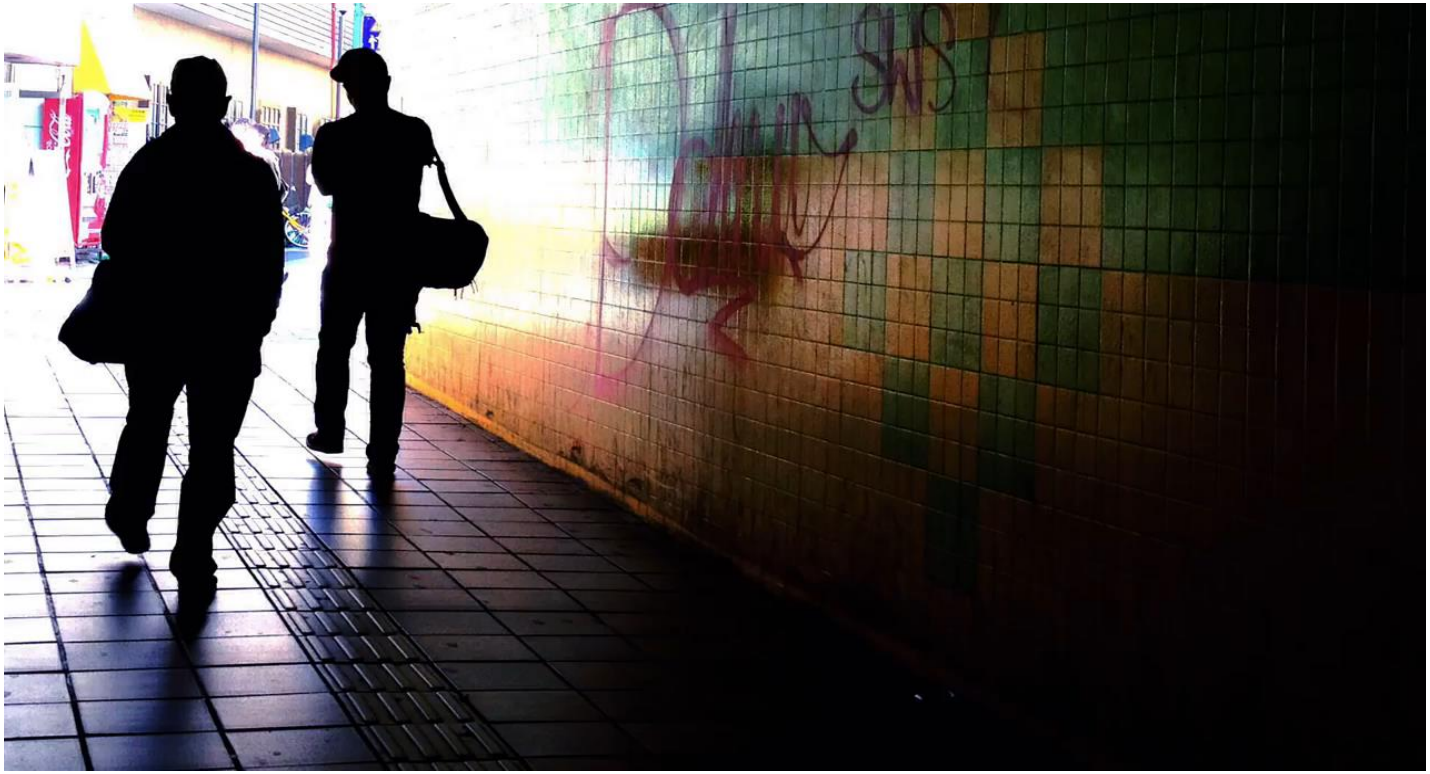
However, if cognitive interviews are neutrally worded, the accuracy of their recall increases.

Darren Matthews/Alamy

When memory fails

When memory fails

- Repressed memories are controversial



https://www.youtube.com/watch?v=D5sk504Yc94

When memory fails

- **Suggestibility**:
  - the development of biased memories from misleading information
    - Different wordings of questions after an event may alter the participants’ memories of the event **->** Loftus
  - Memories can be distorted, or even implanted, with false information
  - Memories can be altered by the intrusion of generic knowledge or semantic associations

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Do you believe you can remember events from your past accurately?

0

Yes

0

No


LECTURE IS LONG, END IT MUST
makeameme.org